

# Guian Fang

WORLD MODELS · VIDEO GENERATION

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“Building AI-powered game worlds indistinguishable from reality.”

## Summary

Ph.D. student at NUS, advised by Prof. Mike Zheng Shou. My research focuses on **Video World Models** and **long-form video generation**, with recent work on **video diffusion distillation** and embodied intelligence, aiming to build interactive, real-time, and physically plausible world simulators. I have published at top venues including ECCV, TNNLS, and LREC-COLING. I also have extensive industry experience collaborating with Huawei Noah's Ark Lab, Shanghai AI Lab, Alibaba DAMO Academy, and several AI startups on production-level generative AI systems.

## Education

### National University of Singapore

PH.D. IN ELECTRICAL AND COMPUTER ENGINEERING

- Advisor: **Prof. Mike Zheng Shou** (Show Lab)
- Research Focus: World Models, Video Generation, Embodied Intelligence

Singapore

Aug. 2024 - Present

### Sun Yat-sen University

B.ENG. IN INTELLIGENT SCIENCE AND TECHNOLOGY

- GPA: **3.7/4.0**, Rank **1st** in final two semesters
- Advisor: **Prof. Xiaodan Liang** (HCP-I2 Lab), co-advised by Prof. Shengcai Liao

Shenzhen, China

Sep. 2020 - Jul. 2024

## Skills

|                    |                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------|
| <b>Research</b>    | Diffusion Models, Video Generation, World Models, RLHF, Large Language Models, 3D Vision   |
| <b>Frameworks</b>  | PyTorch, Diffusers, DeepSpeed, Accelerate, FSDP, JAX/Flax                                  |
| <b>Programming</b> | Python, C/C++, CUDA, LaTeX, Web (HTML, CSS, JavaScript)                                    |
| <b>DevOps</b>      | Git, Linux, Docker, Slurm, Kubernetes (thousand-GPU multi-node training), Weights & Biases |
| <b>Languages</b>   | Chinese (Mandarin, native), English (fluent in reading/writing)                            |

## Experience

### Video World Model for Interactive Storytelling

RESEARCH SCIENTIST

- Building real-time interactive video world models for next-generation AI-powered storytelling

Utopai Studios

Sep. 2024 - Feb. 2026

### Controllable Character Animation

RESEARCH SCIENTIST

- FramePrompt: In-context Controllable Animation with Zero Structural Changes

MikoAI Ltd.

Jul. 2024 - Jun. 2025

### Text-to-3D World & Multimodal Agents

RESEARCH SCIENTIST

- Image-to-3D generation pipeline for production environments

Cybever AI

Sep. 2023 - Jul. 2024

### Generative AI & RLHF for Text-to-Image

RESEARCH INTERN

- HumanRefiner: Benchmarking Abnormal Human Generation and Refining with Coarse-to-fine Pose-Reversible Guidance (**ECCV 2024**)
- RealignDiff: Boosting Text-to-Image Diffusion Model with Coarse-to-fine Semantic Re-alignment (**TNNLS 2023**)

Huawei Noah's Ark Lab

Mar. 2023 - Nov. 2023

### Large Language Models & Controllable Image Generation

RESEARCH INTERN

- **LLaMA-Adapter (5.9k stars)**, **LLaMA2-Accessory (2.8k stars)**
- Improving Compositional Text-to-Image Generation with Large Vision-Language Models

Shanghai AI Lab (OpenGVLab)

Jul. 2023 - Nov. 2023

### Multimodal LLM & Chart Understanding

RESEARCH INTERN

- ChartThinker: A Contextual Chain-of-Thought Approach to Optimized Chart Summarization (**LREC-COLING 2024**)

Alibaba DAMO Academy

Aug. 2023 - Nov. 2023

## Publications

\* Equal contribution.

### CONFERENCE & JOURNAL PAPERS

- Yuchao Gu, **Guian Fang**, Yuxin Jiang, Weijia Mao, Song Han, Han Cai, & Mike Zheng Shou. (2026). **AnyFlow**: Any-Step Video Diffusion Model with On-Policy Flow Map Distillation. *European Conference on Computer Vision (ECCV 2026 Oral)* (with **NVIDIA**).
- Heyuan Gao\*, Bangxun Tang\*, Yiren Song\*, **Guian Fang**, Zijian He, Jie Yang, & Mike Zheng Shou. (2026). **PAI-Studio**: Cinematic Video Background Replacement with Camera-Aware Motion. *European Conference on Computer Vision (ECCV 2026)*.
- **Guian Fang**, Mengsha Liu, & Mike Zheng Shou. (2026). **Declare, Compile, Look**: Coordinate-Free Layout Generation with Vision-in-the-Loop Repair. *ECCV 2026 Multimodal Digital Agents Workshop (MDA 2026 Talk)*.
- **Guian Fang**\*, Wenbiao Yan\*, Yuanfan Guo\*, Jianhua Han, Zutao Jiang, Hang Xu, Shengcai Liao, & Xiaodan Liang. (2024). **HumanRefiner**: Benchmarking Abnormal Human Generation and Refining with Coarse-to-fine Pose-Reversible Guidance. *European Conference on Computer Vision (ECCV 2024)*.
- **Guian Fang**\*, Zutao Jiang\*, Jianhua Han, Guansong Lu, Hang Xu, Shengcai Liao, Xiaojun Chang, & Xiaodan Liang. (2023). **RealignDiff**: Boosting Text-to-Image Diffusion Model with Coarse-to-fine Semantic Re-alignment. *IEEE Transactions on Neural Networks and Learning Systems (TNNLS 2023)*.
- Mengsha Liu, Daoyuan Chen, Yaliang Li, **Guian Fang**, & Ying Shen. (2024). **ChartThinker**: A Contextual Chain-of-Thought Approach to Optimized Chart Summarization. *Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*.
- Ziyang Ma, Mengsha Liu, **Guian Fang**, & Ying Shen. (2025). **LTCR**: Long-Text Chinese Rumor Detection Dataset. *Neural Computing and Applications (NCAA)*.

### PREPRINTS

- **Guian Fang**, Yuchao Gu, & Mike Zheng Shou. (2025). **FramePrompt**: In-context Controllable Animation with Zero Structural Changes. *arXiv preprint*.
- Jay Zhangjie Wu\*, **Guian Fang**\*, Haoning Wu\*, et al. (2024). Towards A Better Metric for Text-to-Video Generation. *arXiv preprint*.
- Jay Zhangjie Wu, **Guian Fang**, et al. (2025). **VEditBench**: Holistic Benchmark for Text-Guided Video Editing. *Preprint*.
- Song Wen, **Guian Fang**, Renrui Zhang, Peng Gao, Hao Dong, & Dimitris Metaxas. (2023). Improving Compositional Text-to-Image Generation with Large Vision-Language Models. *arXiv preprint*.
- Xudong Lu, Renrui Zhang, **Guian Fang**, et al. (2024). **FocusDiff**: Foreground Self-Decoupling Customization for Text-to-Image Generation. *arXiv preprint*.

## Academic Services

- **Conference Reviewer**: CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, AAAI, AISTATS, ACL Rolling Review (ACL, EMNLP, NAACL)
- **Workshop Organizer**: LOVEU @ CVPR 2024 — Long-form Video Understanding Towards Multimodal AI Assistant and Copilot

## Honors & Awards

### COMPETITION

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|------|-----------------------------------------------------------------------------------------|-----------------|
| 2023 | <b>1st Place</b> , 4th National College Computer Ability Challenge                      | Shenzhen, China |
| 2022 | <b>2nd Place</b> , Asia and Pacific Mathematical Contest in Modeling                    | Beijing, China  |
| 2022 | <b>Silver Medal</b> , China Collegiate Algorithm Design & Programming Challenge Contest | Shanghai, China |
| 2022 | <b>2nd Place</b> , The 2022 Social Computing Innovation Competition                     | Hangzhou, China |

### SCHOLARSHIP

|      |                                                                                                 |                 |
|------|-------------------------------------------------------------------------------------------------|-----------------|
| 2023 | <b>1st Place</b> , National Scholarship (Rank <b>1st</b> )                                      | Shenzhen, China |
| 2023 | <b>1st Place</b> , SYSU Outstanding Student Scholarship (Top <b>5%</b> )                        | Shenzhen, China |
| 2023 | <b>1st Place</b> , Li Xuerou Fund Scholarship (Among the top <b>10</b> students in the college) | Shenzhen, China |
| 2022 | <b>Recipient</b> , Huawei Intelligent Foundation Scholarship                                    | Shenzhen, China |
| 2022 | <b>Recipient</b> , National Encouragement Scholarship                                           | Shenzhen, China |